

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: tsplib_no_replay.
- Best TSPLIB holdout gap: tsplib_no_replay.
- Best synthetic holdout gap: tsplib_no_replay.

Run Metadata

- run_name: run_20260513_131240_d
- started_at_local: 2026-05-13 13:12:40
- finished_at_local: 2026-05-13 13:22:31
- duration_hhmm: 00:10
- duration_seconds: 590.409
- seed_offset: 3000
- replicate_label: d
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
tsplib_no_replay	none	score_only	False	0.213387	0.064916	0.159398	0.80218	0.56	-0.031047
tsplib_random_replay	random	score_only	False	0.213387	0.064916	0.159398	0.823575	0.56	-0.03024
tsplib_failure_replay	failure	score_only	False	0.213387	0.064916	0.159398	0.853124	0.56	-0.029193
tsplib_failure_replay_compression	failure	novelty_gate	True	0.213387	0.064916	0.159398	0.0	0.56	0.0

Condition Notes

tsplib_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.

- Accepted-epoch count 3, mean accepted code novelty 0.80218, and final complexity 0.56.
- Adaptation efficiency -0.031047 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.823575, and final complexity 0.56.
- Adaptation efficiency -0.03024 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.853124, and final complexity 0.56.
- Adaptation efficiency -0.029193 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.56.

- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
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Judge Appendix

Summary of Results for Replay-Aware TSP Benchmark Suite

Condition	Replay Mode	Selection Mode	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap	Mean Code Novelty	Compression Pressure	Comment
***tsplib_no_replay**	none	score_only	0.213387	0.064916	0.159398	0.80218	false	Best overall transfer and gaps
tsplib_random_replay	random	score_only	0.213387	0.064916	0.159398	0.823575	false	Same gaps as no_replay, higher novelty
tsplib_failure_replay	failure	score_only	0.213387	0.064916	0.159398	0.853124	false	Same transfer/gaps, higher novelty
tsplib_failure_replay_compression	failure + compression	novelty_gate	0.213387	0.064916	0.159398	0.0	true	Equal gaps but zero code novelty, compression pressure

Interpretation

- **Transfer Performance (Key metric):**

All conditions exhibit identical mean TSPLIB holdout gaps (0.2134), synthetic holdout gaps (0.0649), and transfer gaps (0.1594). This indicates **no measurable difference in transfer ability between replay modes** (no replay, random replay, failure replay, or compression-aware replay).

- **Code Novelty vs Transfer:**

Despite identical gaps, conditions with random replay and failure replay show **higher code novelty (~0.82-0.85)** compared to no replay (~0.80) and zero novelty under compression-aware failure replay. Therefore, **code novelty does not correlate with improved transfer or optimality gaps here.**

- **Compression Pressure:**

The failure replay with compression mode shows zero code novelty yet no decline in transfer or gap metrics. This suggests the compression impacts code novelty significantly but **without harming final transfer or solution quality**.

- **Replay Mode Impact on Optimality Gap:**

No replay achieves the best transfer and holdout gaps, and replay conditions (random or failure) do not improve or degrade these gaps. Compression-aware replay does not affect gaps either.

- **Behavior and Complexity:**

Across all conditions, complexity and behavior profiles remain stable (complexity ~0.56, balanced behavior).

Conservative Conclusions

- **Best transfer and gap metrics are achieved under the no replay condition (tsplib_no_replay).**
- **Replay modes (random or failure) do not improve transfer or reduce optimality gaps despite higher code novelty.**
- **Compression-aware failure replay removes code novelty but maintains equal transfer and optimality performance.**
- **Lexical or code novelty increases observed under replay do not correspond to algorithmic improvements in gap or transfer metrics.**
- **No empirical evidence supports that replay (random or failure) or compression-aware replay improves optimality-gap or transfer beyond no replay baseline in this suite.**